Learned Image Compression for Machine Vision: A Multi-Task Evaluation of Fidelity, Classification, Detection, and OCR

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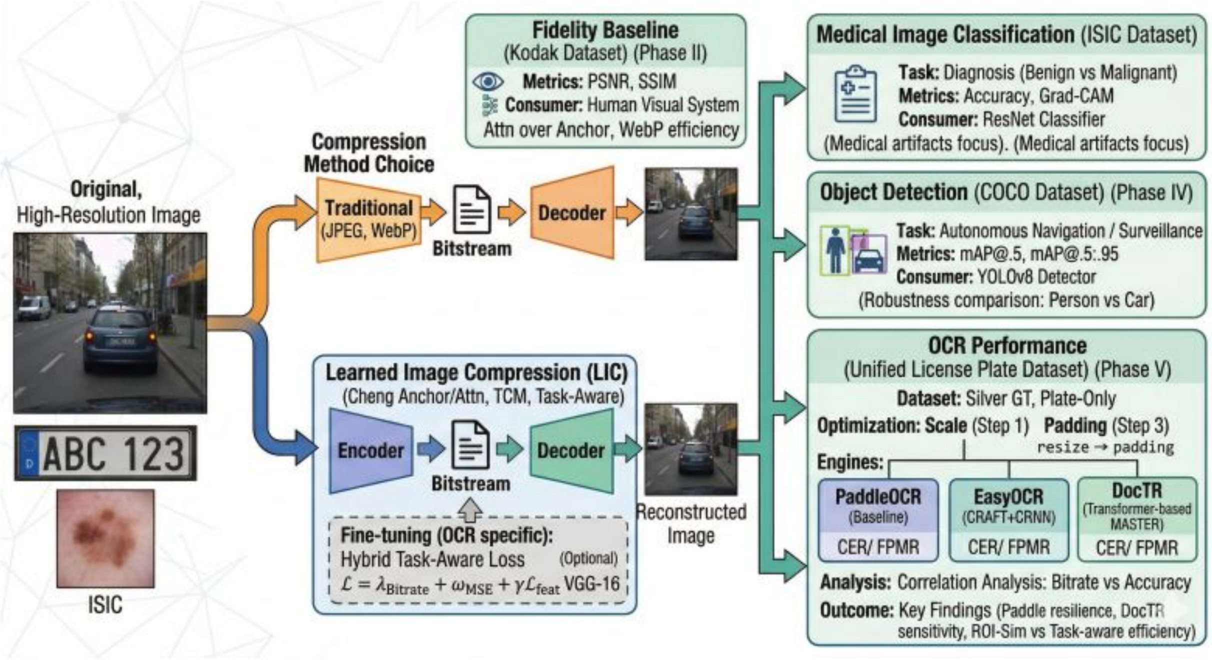
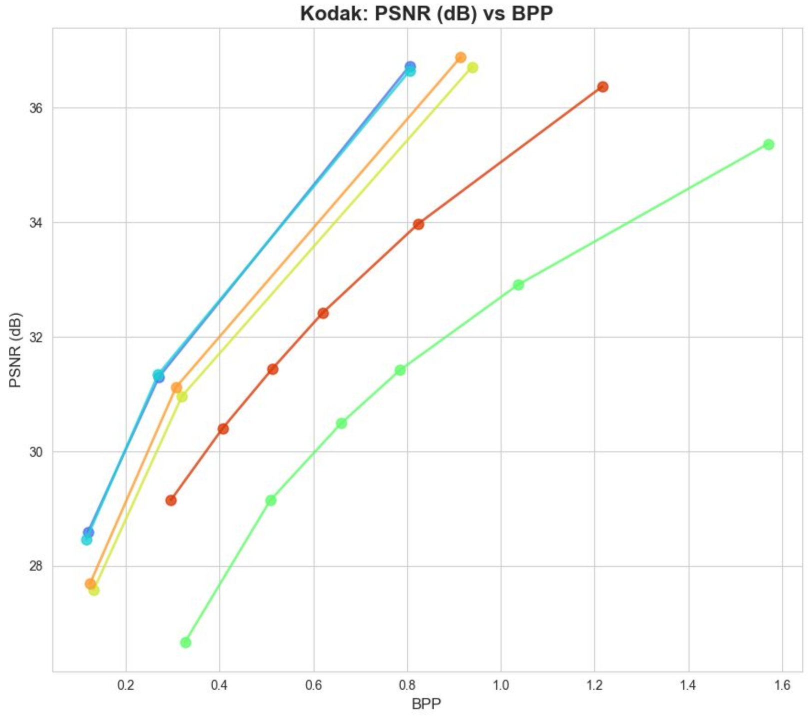
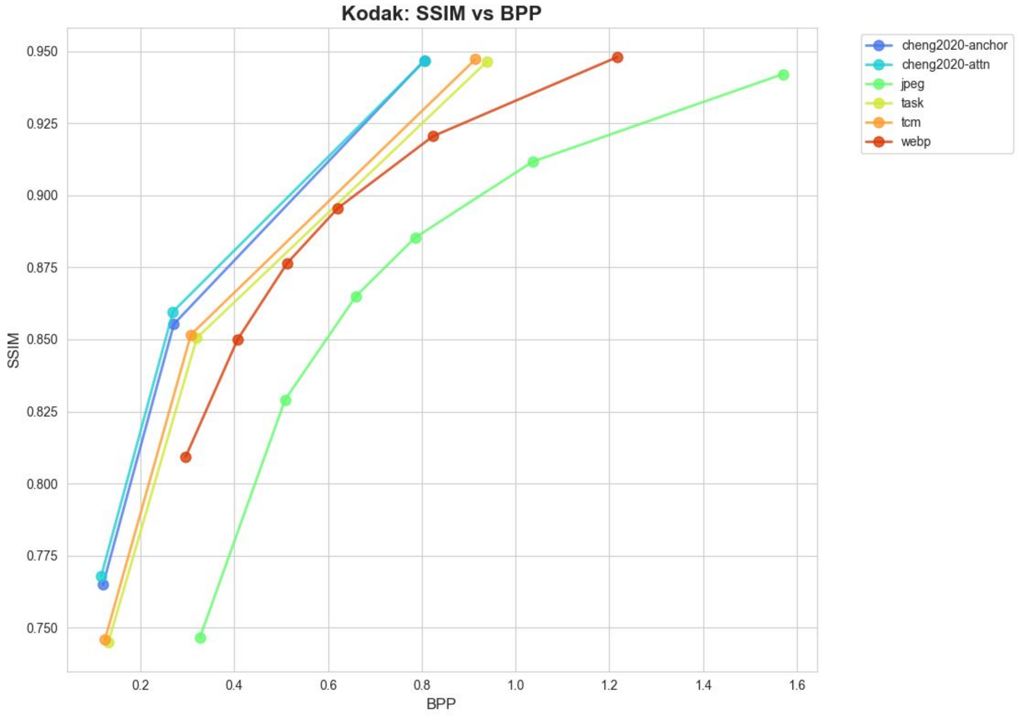
The essence of work



Fidelity



Kodak: PSNR (dB) vs BPP
PSNR (dB)
36
34
32
30
28
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
BPP



Kodak: SSIM vs BPP
SSIM
0.950
0.925
0.900
0.875
0.850
0.825
0.800
0.775
0.750
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
BPP
cheng2020-anchor
cheng2020-attn
jpeg
task
tcm
webp

Fidelity

ORIGINAL
kodim04.png

JPEG (Q15)
BPP: 0.337 | PSNR: 29.33

WebP (Q15)
BPP: 0.224 | PSNR: 30.41

Cheng Anchor (Q3)
BPP: 0.169 | PSNR: 32.11

Cheng Attention (Q3)
BPP: 0.169 | PSNR: 32.26

TCM Proxy (Q3)
BPP: 0.202 | PSNR: 32.00

Task-Aware (Q3)
BPP: 0.219 | PSNR: 31.90

Classification

- **Task:** Skin lesion classification
- **Dataset:** ISIC 2019 dataset
- **Main metric:** AUROC
- **Supporting metrics:** Accuracy, Macro F1, Macro Recall
- **Efficiency metric:** BEI_norm, combining performance with bitrate
- **Goal:** test whether compressed images still preserve diagnostic information

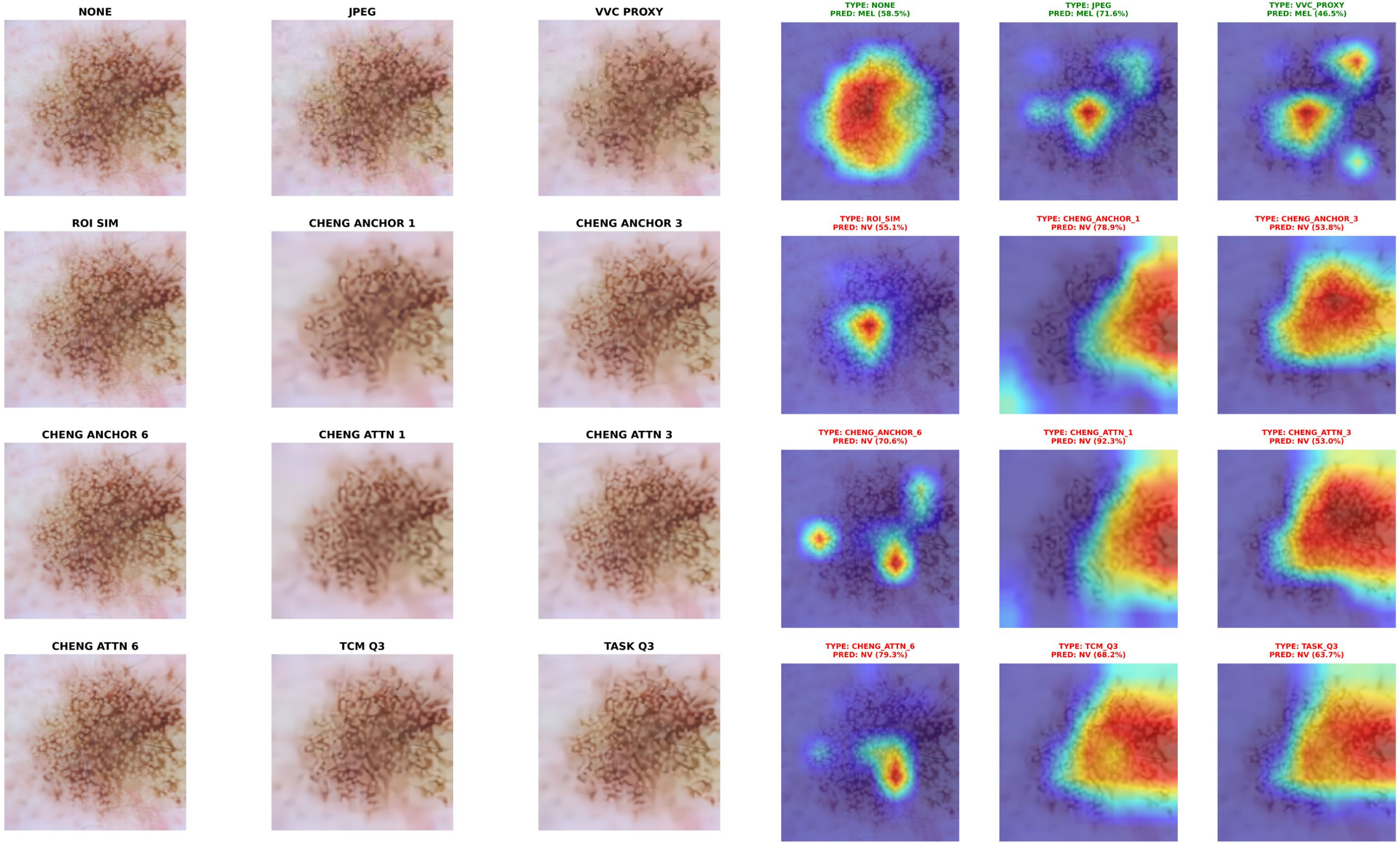
Classification. Results

Compression	Avg BPP	AUROC	Accuracy (%)	F1 (Macro)	Recall (Macro)	BEI_norm*
none	9.9977	0.9012	65.27	0.4522	0.4541	0.3315
Jpeg_80	0.4373	0.7850	53.18	0.3009	0.3143	0.7514
roi_sim	0.7378	0.7731	49.49	0.2892	0.3025	0.7181
cheng_attn_6	0.2465	0.7649	49.56	0.3005	0.322	0.7463
vvc_proxy	0.1955	0.7637	49.43	0.2774	0.3018	0.7489
cheng_anchor_6	0.2450	0.7614	49.26	0.2982	0.3207	0.7429
tcm_q3	0.0927	0.7004	44.76	0.2408	0.2682	0.6939
task_q3	0.1117	0.6936	44.05	0.2368	0.2650	0.6859
cheng_attn_3	0.0707	0.6814	40.26	0.2371	0.2662	0.6766
cheng_anchor_3	0.0696	0.6744	39.37	0.2161	0.2473	0.6697
cheng_attn_1	0.0321	0.6567	38.16	0.1966	0.2296	0.6546
cheng_anchor_1	0.032	0.6382	36.37	0.1815	0.2172	0.6361

Figure. Average (by Classifier) performance evaluated on test dataset compressed by different models

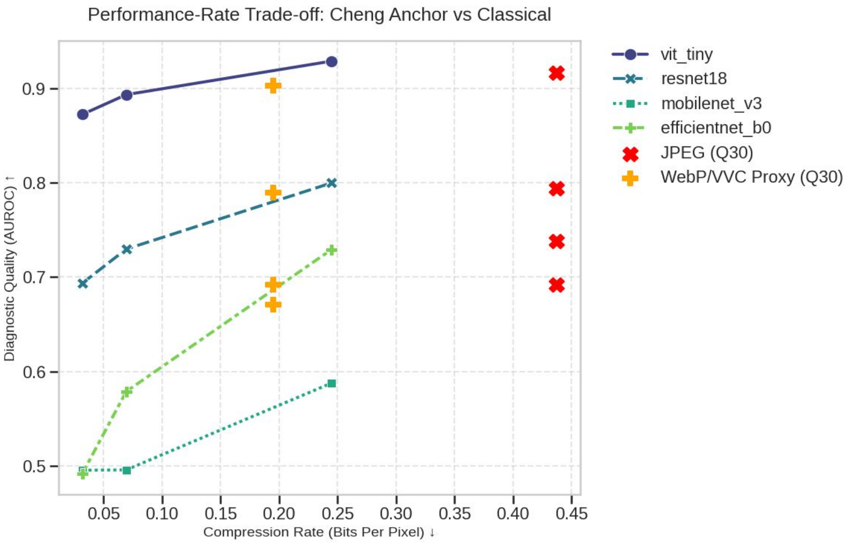
* – BEI_norm = AUROC × $\exp\{-\alpha(BPP/BPP_{max})\}$, $\alpha=1$

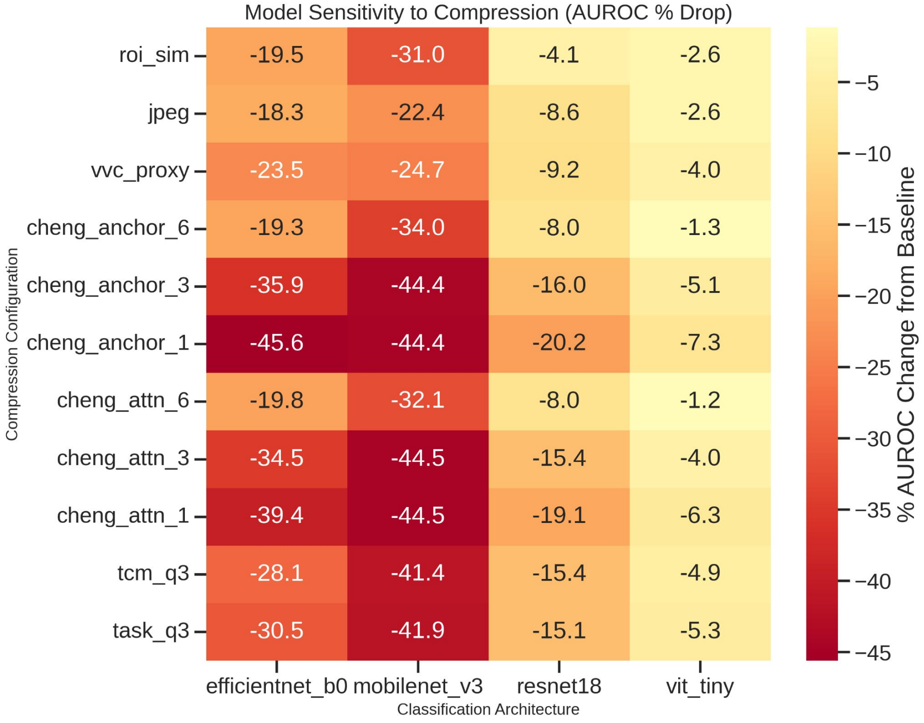
Classification. XAI



Classification. Trade-off

- Compression causes a **consistent AUROC drop** across all architectures
- **Lightweight models (EfficientNet-B0, MobileNetV3)** are more sensitive to bitrate reduction
- **ResNet18 and ViT-Tiny** show higher robustness, especially at moderate compression
- Moderate compression (≈0.2–0.25 BPP) preserves most diagnostic signal
- Very aggressive compression (<0.1 BPP) leads to sharp performance degradation





Classification

No compression gives the best raw performance: AUROC **0.9012**, Accuracy **65.27%**

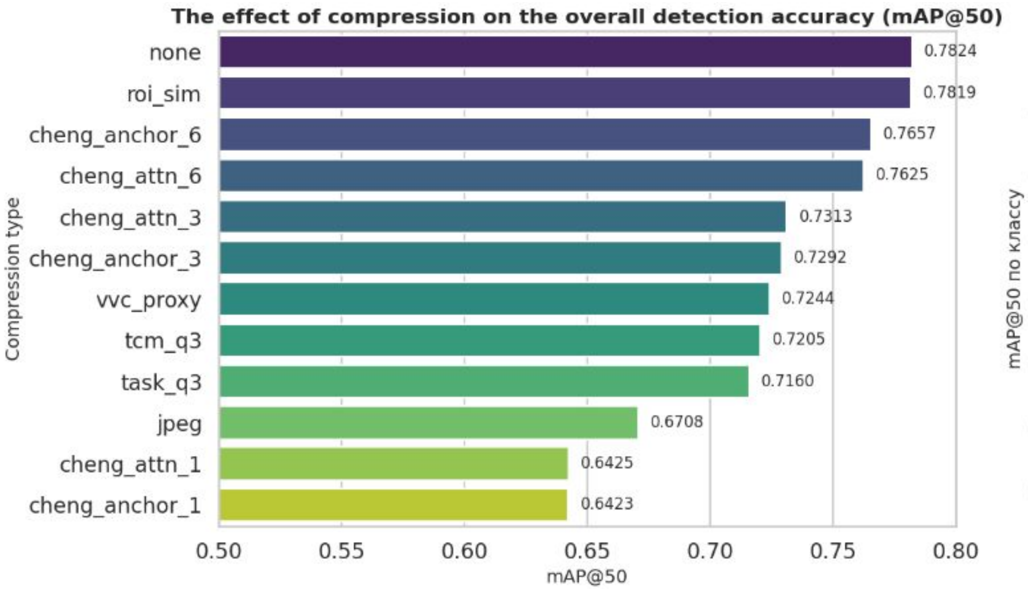
But it is the **least efficient**: BEI_norm **0.3315**

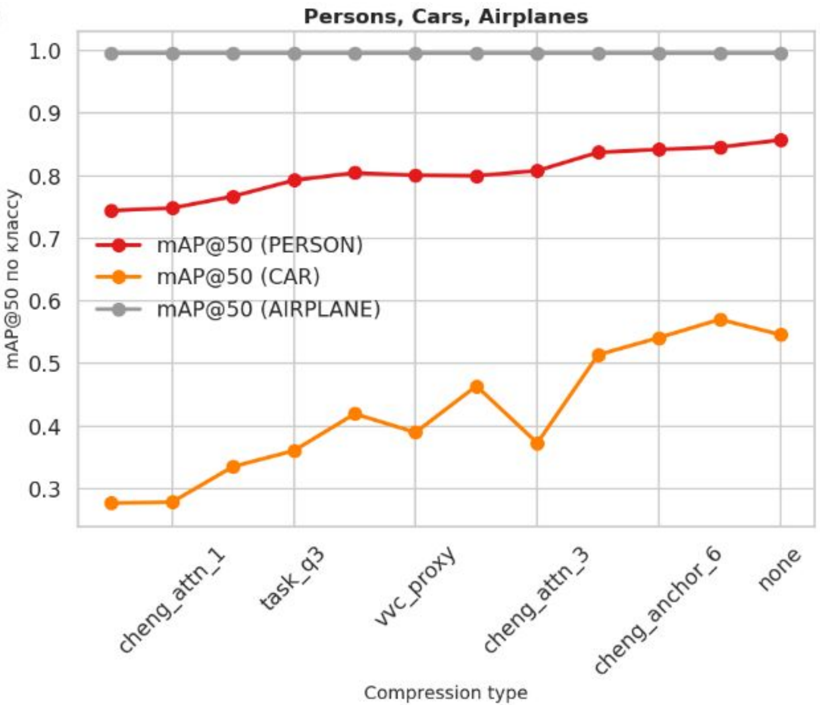
Strong compressed options still keep useful performance:

- **JPEG (Quality 80):** AUROC **0.7850**, BEI_norm **0.7514**
- **VVC proxy:** AUROC **0.7637**, BEI_norm **0.7489**
- **Cheng attention 6:** AUROC **0.7649**, BEI_norm **0.7463**

Very aggressive compression causes a clear drop in classification quality

Object Detection

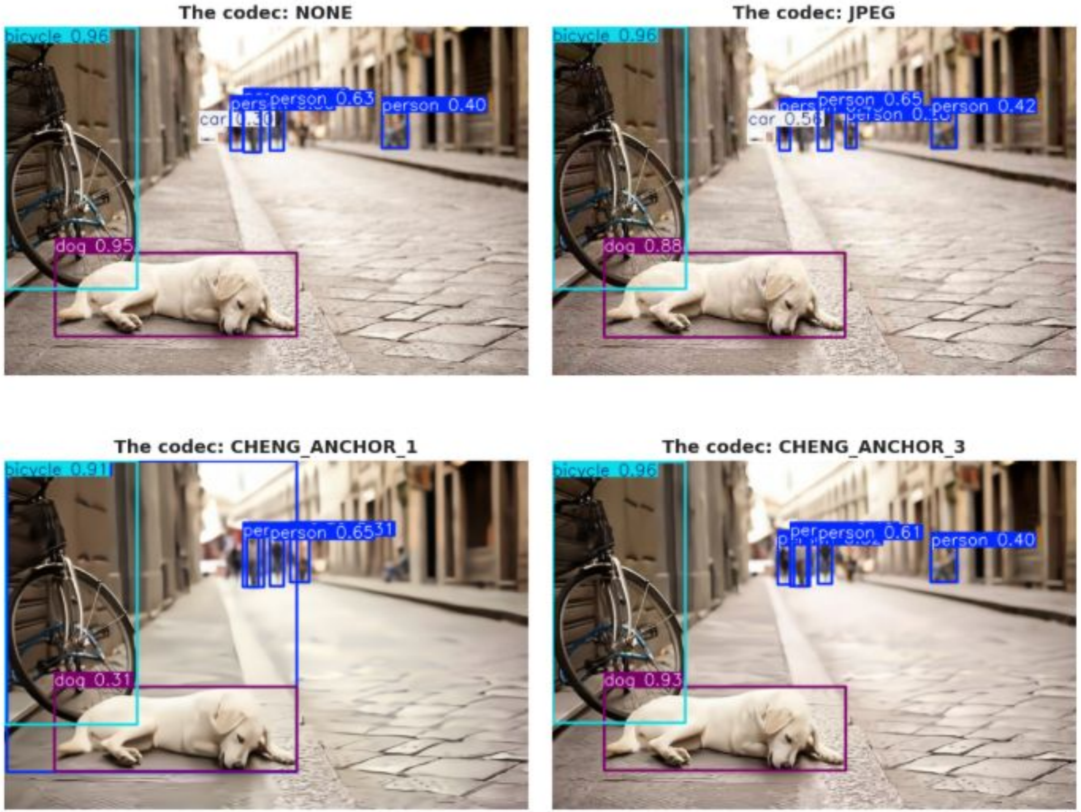




Example

Less quality

Less confidence
&
loss of objects



Object Character Recognition

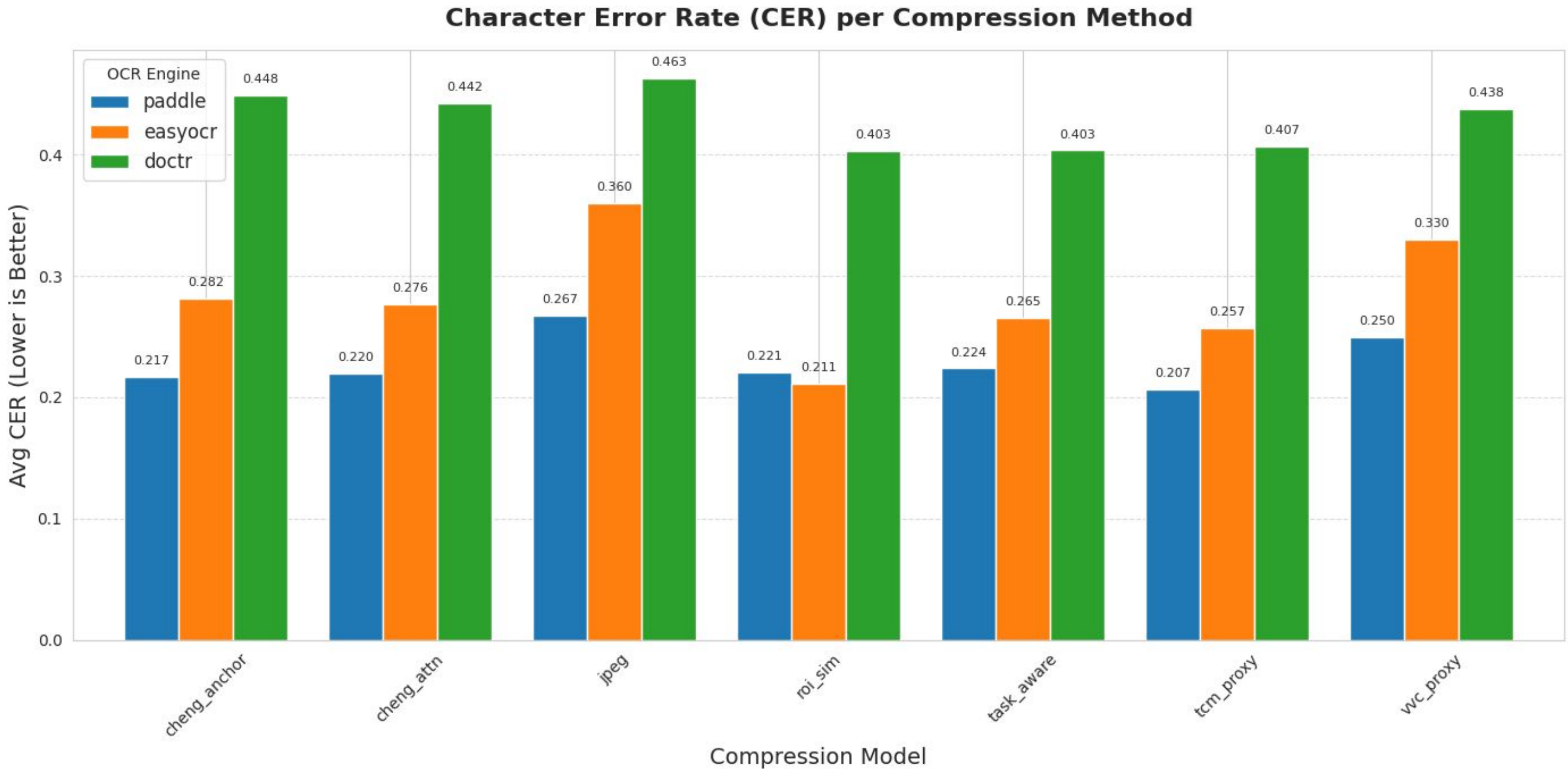
The problem:

ORIGINAL (Index 255)
OCR: HR26BC55I4

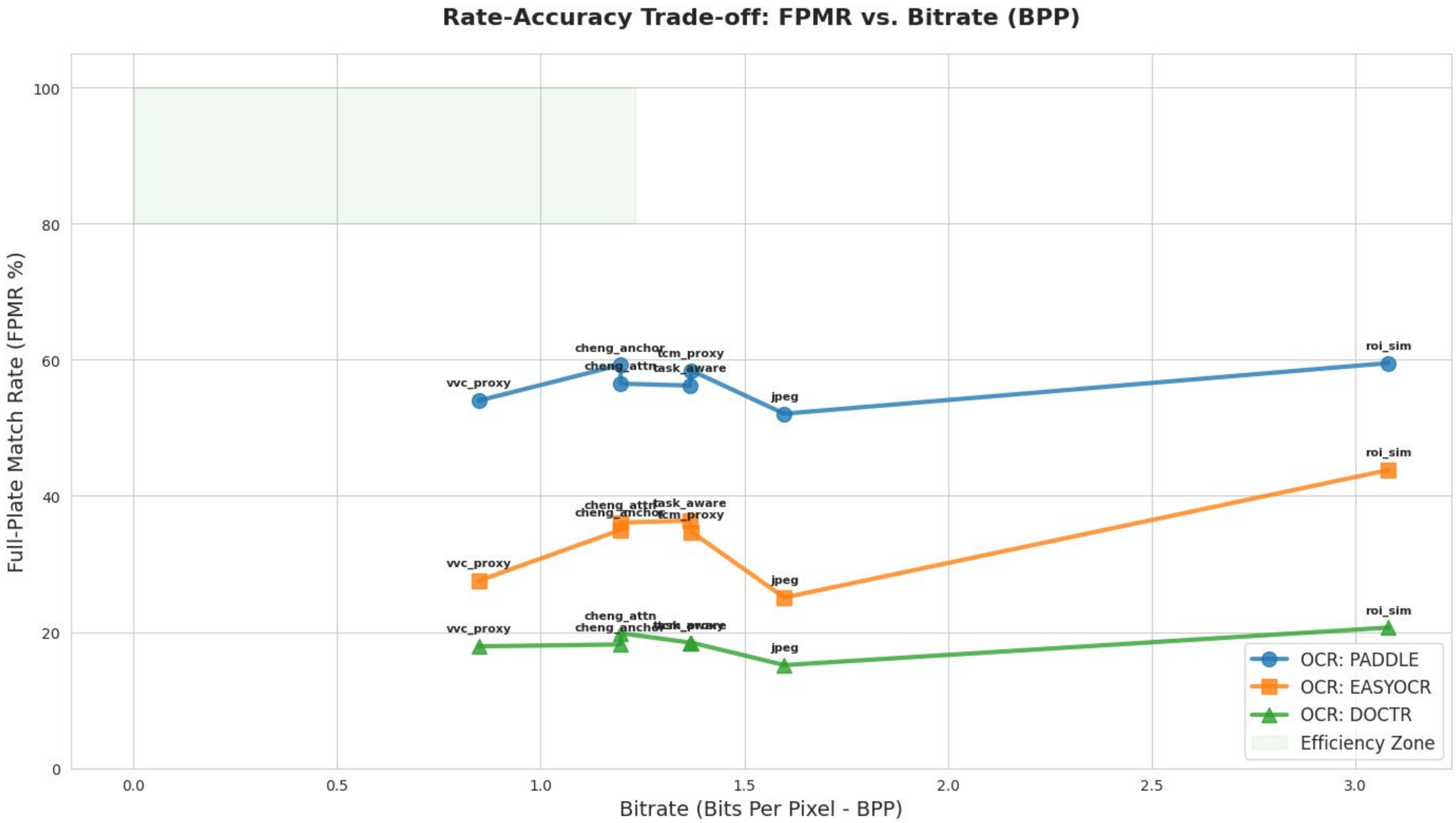
TCM_PROXY COMPRESSED
OCR: HR26BC5514

- **Task:** Object character recognition
- **Dataset:** Roboflow License Plates + Kaggle Car License Plate Detection
- **Main metrics:** average CER
- **Efficiency metric:** combining performance (accuracy, full plate match rate) with bitrate
- **Goal:** test whether the OCR engine recognizes the same text in compressed and uncompressed images (true label = label returned by the OCR engine for the uncompressed image)

Object Character Recognition results: CER comparison



Object Character Recognition results: BPP vs FPMR (accuracy)



Object Character Recognition: summary

- PaddleOCR: Remains the clear leader. It is remarkably resilient to "neural smoothing."
- The neural models (cheng, tcm, task_aware): lower bitrate (~1.2–1.3 BPP) compared to JPEG (1.6 BPP), but higher accuracy across all engines.
- tcm_proxy and cheng_anchor achieve nearly identical accuracy to roi_sim on PaddleOCR with 2 times smaller BPP.
- vvc_proxy (modern non-learned method) achieves the lowest bitrate in the entire study (0.84 BPP) and still manages to outperform JPEG (e.g., 53.9% vs 52.1% FPMR on Paddle).

Conclusions:

1. Paradigm Shift Confirmed: Learned Image Compression (LIC) architectures provide superior foundation for machine-vision-centric data representation compared to classical codecs (JPEG, WebP)
2. Fidelity vs. Task Utility:
    - LIC models create different artifact types than JPEG
    - LIC smooths high-frequency details vs. JPEG's blockiness
    - This fundamentally impacts downstream task performance
3. Multi-Task Evaluation Success:
    - ✓ Fidelity baseline established (Kodak dataset)
    - ✓ Medical imaging reliability quantified (ISIC dataset)
    - ✓ Object detection robustness measured (YOLOv8)
    - ✓ OCR performance assessed with "Efficiency Zone" defined
4. Architecture Matters: For the studied datasets, the basic convolutional codecs proved to be sufficient; the attention mechanisms did not reveal significant spatial dependencies that could improve compression.

Future Work:

1. Real-time Inference Constraints:
    - Measure decoding speed on embedded hardware
    - Evaluate memory footprint requirements
2. Extended Medical Studies:
    - Focus on class-imbalanced medical datasets
    - Investigate heightened artifact sensitivity in rare conditions

Thank you for your attention!